SATYAM KUMAR

Kolkata, West Bengal | +91-7209001826 | sa.25.07.2006@gmail.com
linkedin.com/in/satyam-kumar-006b52324 | github.com/satyam-257

PROFESSIONAL SUMMARY

B.Tech CSE (Data Science) student who builds ML systems end to end: a leakage-aware GBDT stacking pipeline for a Kaggle Playground competition (0.968 CV ROC-AUC), an Indian Railways ETA model that cuts error 42% over the timetable baseline, and an ML experiment-analysis platform with verified metrics and calibration checks. Also ships full-stack products in React/TypeScript and Python. Seeking ML / AI / Data Science internships.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, TypeScript
Machine Learning: scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, SciPy, Random Forest, K-Means
ML Techniques: Feature engineering, out-of-fold target encoding, stratified K-fold CV, stacking, hyperparameter tuning, leakage audits
Deep Learning: PyTorch, TorchScript, transfer learning (fine-tuned MobileNetV3-Small)
Data & GenAI: Pandas, NumPy, Matplotlib, Seaborn, OpenAI and Gemini APIs (LLM insight synthesis), FastAPI, REST APIs
Tools: Git, GitHub, GitHub Actions, Jupyter, Google Colab, Kaggle, React, Vite, Supabase

MACHINE LEARNING & AI PROJECTS

TrackX (formerly RailSight AI) | *Dynamic Train ETA Prediction for Indian Railways*
Python, FastAPI, XGBoost, scikit-learn, Open-Meteo, React, TypeScript github.com/satyam-257/rail_collector

- Built a 5-source data pipeline (historical delays, route/station GeoJSON, station master, Open-Meteo weather, RailRadar client) with leakage-free delay features computed on training splits only.
- Modelled ETA as delay deviation from the timetable; on a journey-aware held-out split (515 snapshots, prototype dataset) XGBoost reached 10.37 min MAE vs 17.97 for the schedule baseline (-42%); compared with Random Forest (10.62) and a delay-risk classifier.
- Served ETA, feature-attribution and congestion endpoints via FastAPI; built a React/TypeScript dashboard (network map, live tracking, model-comparison charts).

ExperimentPilot | *ML Experiment Tracking and Model-Behavior Analysis Platform*
Python, FastAPI, SQLite, Pydantic, React, TypeScript, Vite, Gemini / OpenAI APIs github.com/satyam-257/ExperimentPilotAI

- Built a model-agnostic platform to log experiment runs and independently recompute accuracy, precision, recall, macro-F1, confusion matrix, 10-bin Expected Calibration Error and regression metrics (MAE, RMSE, R^2) from sample-level predictions, separating verified from author-reported numbers.
- Added comparison guardrails that block runs with mismatched datasets, splits or task types, plus hyperparameter diffs and a false-positive / false-negative / disagreement explorer (demo data: synthetic CT-hemorrhage and credit-risk fixtures).
- Implemented an offline rule-based insight engine with optional Gemini / OpenAI synthesis, grouping findings into measurements, hypotheses and recommendations; FastAPI + SQLite backend with tests, Render / Vercel deployment config.

Traverly | *Domestic Travel Planning Platform*
React, TypeScript, Vite, Tailwind, Supabase, FastAPI, pandas github.com/satyam-257/Traverly---you-travel-partner

- Built a full-stack platform (trip planner, budget estimator, transport booking, meetups) with Supabase auth/database and GitHub Actions deployment.
- Implemented a FastAPI budget-estimation service: pandas lookups over India city-cost datasets, fuzzy destination matching and a multi-factor cost model (travel style, season, party size, transport) returning itemised estimates and day-by-day tips.

KAGGLE COMPETITIONS & MACHINE LEARNING EXPERIMENTS

Predicting Smartphone Addiction | *Playground Series S6E8 (participation; binary classification, ROC-AUC)* Competition · Code

- Engineered 172 features, including leakage-free 10-fold out-of-fold target encoding (m-estimate smoothing) of numeric columns treated as discrete grid values, plus digit, trigonometric, interaction, cohort-aggregate and missingness features.
- Trained LightGBM, XGBoost (GPU) and CatBoost (GPU) models and an XGBoost specialist under 10-fold stratified CV, tuned with Optuna; blended percentile-ranked OOF predictions with Nelder-Mead weight optimisation: 0.968 CV ROC-AUC vs 0.957 for a raw-feature CatBoost baseline.
- Logged 28 experiments with keep/reject ablations and investigated the CV-to-leaderboard gap using adversarial validation and target-encoding leakage analysis.

RSNA Knee Abnormality Detection | *Participation (no leaderboard result claimed)* Competition

- [ADD: your verified approach, model and result]

Memecoins: Pump or Dump? | *Participation (no leaderboard result claimed)* Competition

- [ADD: your verified approach, model and result]

EDUCATION

Netaji Subhash Engineering College, Kolkata, West Bengal | B.Tech CSE (Data Science), CGPA 7.6 — 2024 - 2028
Gossner College, Ranchi, Jharkhand | Intermediate (BSEB), PCM, 76% — 2021 - 2023

RELEVANT ACHIEVEMENTS

- Traverly selected for Amazon AI Summit 2026 as one of a limited set of student-built AI projects.
- Campus Ambassador, KSHITIJ (IIT Kharagpur), 2025 (recruited 30+ participants); 100+ CodeChef and 80+ LeetCode problems solved.